

Neural–mechanistic hybrids and physics-informed neural networks for a WNT–RA–HOX model of colorectal cancer stemness

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01 Model & formulation

From signaling to stemness. Seven ODEs connect WNT, APC and retinoic-acid regulation. We study state reconstruction, parameter recovery and learned regulation.

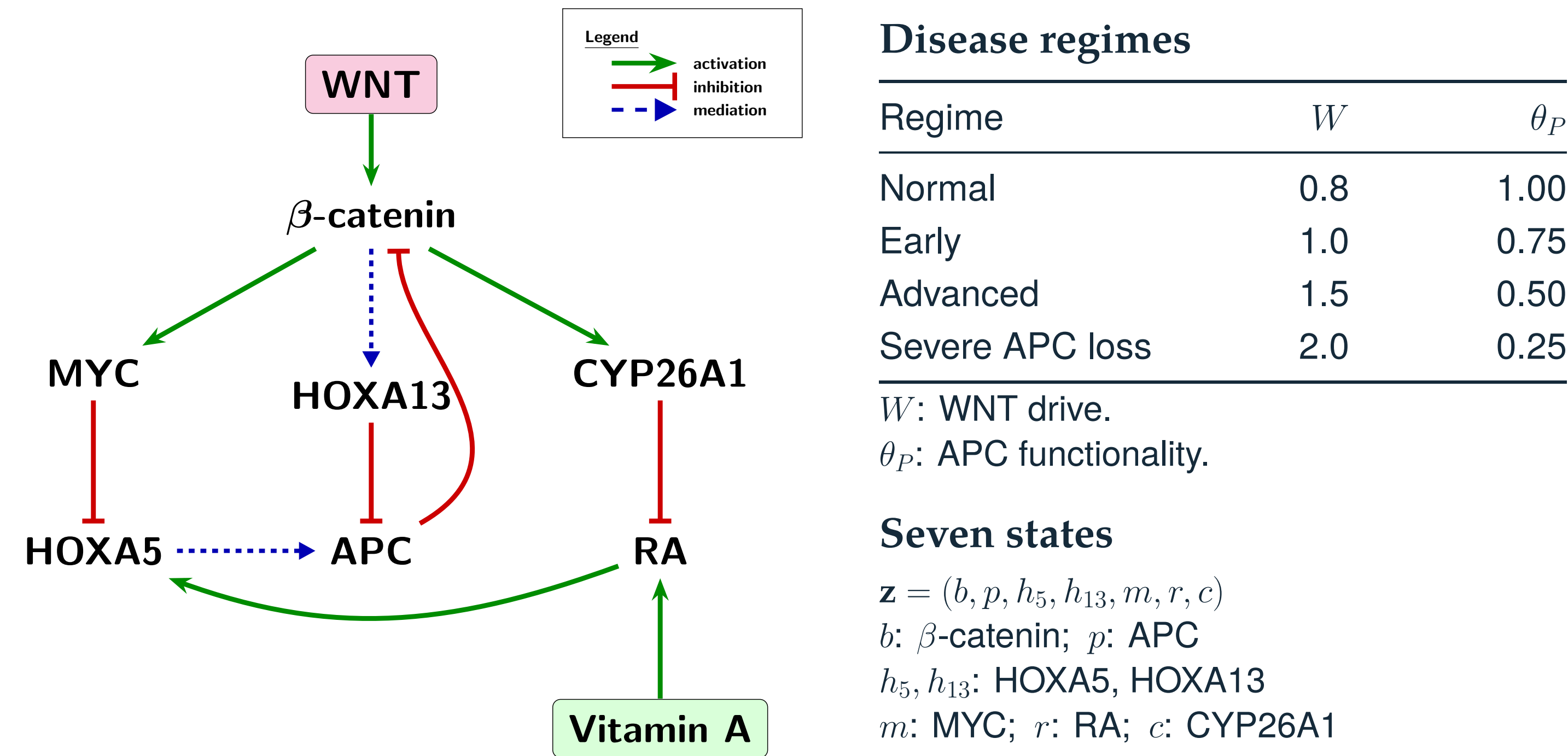


Fig. 1. Regulatory network. Green: activation; red: inhibition; blue dashed: mediation. Early and Advanced denote adenoma.

Nondimensionalization

Scale concentrations by characteristic values and time by β -catenin turnover, $d_B = 1 \text{ h}^{-1}$:

$$X = X_* x, \quad \tau = d_B t, \quad \varepsilon_X = \frac{d_B}{d_X}. \quad (1)$$

Hill activation: $H_n(u; \kappa) = u^n / (\kappa^n + u^n)$. A dot below denotes $d/d\tau$.

$$\begin{aligned} \dot{b} &= W + \eta_{13} H_{n_H}(h_{13}; \kappa_{13}) - b - \lambda_P p b - \lambda_5 \frac{h_5 b}{\kappa_5 + b}, \\ \varepsilon_P \dot{p} &= \frac{1 + \rho_5 h_5}{1 + \rho_B b + \rho_{13} h_{13}} - D_P(\theta_P) p, \\ \varepsilon_5 \dot{h}_5 &= a_5 + \eta_R H_1(r; \kappa_R) - h_5 - \eta_M \frac{m h_5}{\kappa_M + m}, \\ \varepsilon_{13} \dot{h}_{13} &= a_{13} + \eta_{B13} H_{n_B}(b; \kappa_{B13}) + \eta_{M13} H_{n_M}(m; \kappa_{M13}) - h_{13}, \\ \varepsilon_M \dot{m} &= a_M + \eta_{BM} H_{n_B}(b; \kappa_{BM}) - m, \\ \varepsilon_R \dot{r} &= \mu_R(\tau) - r - \lambda_{CC} r, \\ \varepsilon_C \dot{c} &= a_C + \eta_{RC} H_1(r; \kappa_{RC}) + \eta_{BC} H_{n_B}(b; \kappa_{BC}) - c. \end{aligned} \quad (2)$$

APC loss and retinoid forcing

$$\begin{aligned} D_P(\theta_P) &= 1 + \delta_{P1}(1 - \theta_P), \\ \mu_R(\tau) &= \mu_0 + A_R \left[1 + \cos\left(\frac{2\pi\tau}{T_R} - \varphi\right) \right] \\ &\quad + \frac{D_R}{2} [\tanh q(\tau - \tau_1) - \tanh q(\tau - \tau_2)]. \end{aligned} \quad (3)$$

Reference: Radau on $\tau \in [0, 150]$; ATRA pulse [40, 88]. Relative timescales $\varepsilon_R = 0.40$, $\varepsilon_M = 0.60$ produce mild stiffness.

Derived readout: stemness

$$S(\tau) = \frac{b(1 + \alpha_{13} h_{13})}{(1 + p)(1 + \alpha_5 h_5)}, \quad \alpha_5 = \alpha_{13} = 1.$$

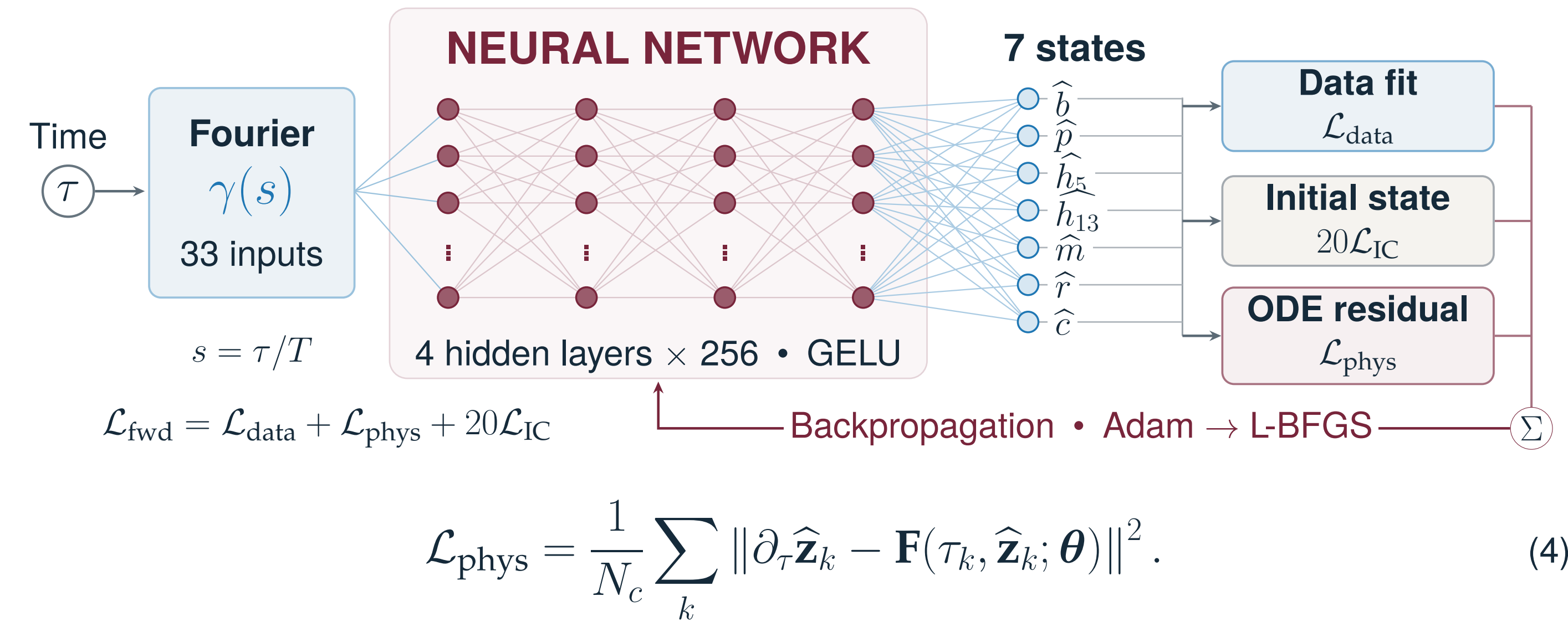
Higher S represents greater stemness in this model.

Scope and next step

Synthetic-data study. Intervention screens fit one equation to exact states with two starts; the constraint screen uses one. Basal recovery can improve while function recovery worsens.³ Next: noisy, sparse-data validation.

02 PINNs & identifiability

Forward problem. Fourier features resolve circadian forcing and the ATRA pulse.¹



Data and initial-state losses are squared errors; physics derivatives use automatic differentiation.

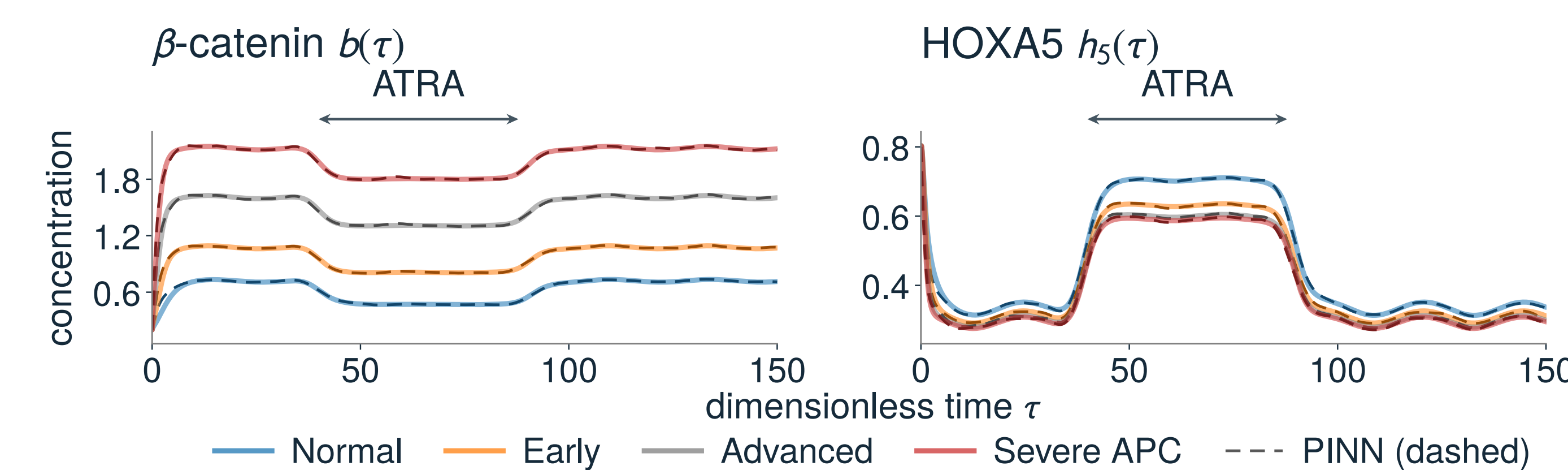


Fig. 2. β -catenin and HOXA5, all four regimes. Solid: Radau; dashed: PINN (40 observations).

Forward error (mean relative L^2): **1.06 %** supervised / 100 labels; **2.41 %** PINN / 40 labels. Means span 7 states and 4 regimes; separate formulations.

Recovering the 36 model parameters

One state network per condition; 36 shared model parameters:

$$\mathbf{r}_k = \hat{\mathbf{z}}_{k+1} - \hat{\mathbf{z}}_k - \frac{1}{2} \Delta \tau_k (\mathbf{F}_k + \mathbf{F}_{k+1}). \quad (5)$$

Trapezoidal residual: no time derivatives; relative state weights; fixed collocation; two starts.

Method	Normal	Early	Advanced	Severe	Total
Autodiff	16	9	6	6	37
Integral	17	16	10	7	50

Table 1. Parameters within 10%; 10 WNT/MYC conditions; 144 parameter–regime pairs. Gain reflects the full integral implementation.

Bayesian geometry and local information

Frozen-state HMC (36 parameters; integral likelihood): $D_P - 1 = \delta_{P1}(1 - \theta_P) \approx \text{constant}$.

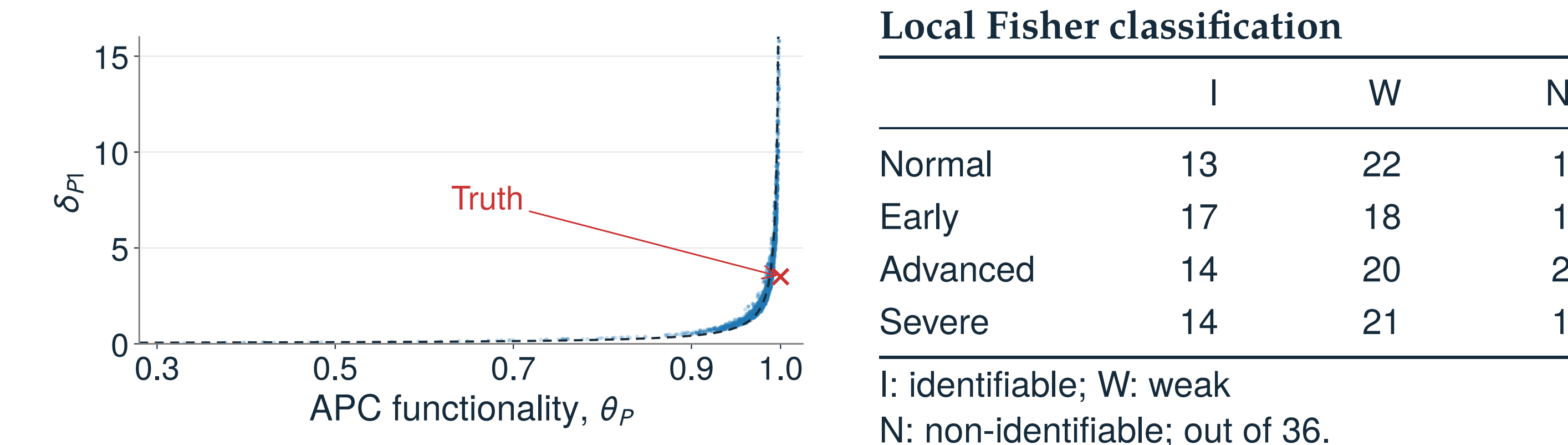


Fig. 3. Normal-regime APC trade-off. At $\theta_P = 1$, δ_{P1} drops out.

Limits. HMC fails ESS/coverage checks; intervals are unreliable. Fisher counts are local. High-WNT saturation weakens APC identifiability.

Key references

- [1] M. Raissi, P. Perdikaris and G. E. Karniadakis. Physics-informed neural networks. *J. Comput. Phys.* 378, 686–707 (2019). doi:10.1016/j.jcp.2018.10.045.
- [2] C. Rackauckas et al. *Universal differential equations for scientific machine learning*. arXiv:2001.04385 (2020).
- [3] T. E. Loman and R. E. Baker. *Functional and parametric identifiability for universal differential equations applied to chemical reaction networks*. arXiv:2510.14140 (2025).

03 Neural–mechanistic hybrid

Learn one relationship within the ODE model. A shared 2×5 tanh network replaces a regulatory term; the remaining biology stays mechanistic.²

$$\begin{aligned} \eta_{BM} H_{n_B}(b; \kappa_{BM}) &\longrightarrow f_\phi(b), \quad f_\phi(0) = 0, \\ \varepsilon_M \dot{m} &= a_M + f_\phi(b) - m. \end{aligned} \quad (6)$$

For MYC, (η_{BM}, κ_{BM}) leave the inverse parameter set: $36 \rightarrow 34$. State networks, remaining parameters and f_ϕ are fitted jointly with the integral residual; neural L2 penalty 10^{-8} .

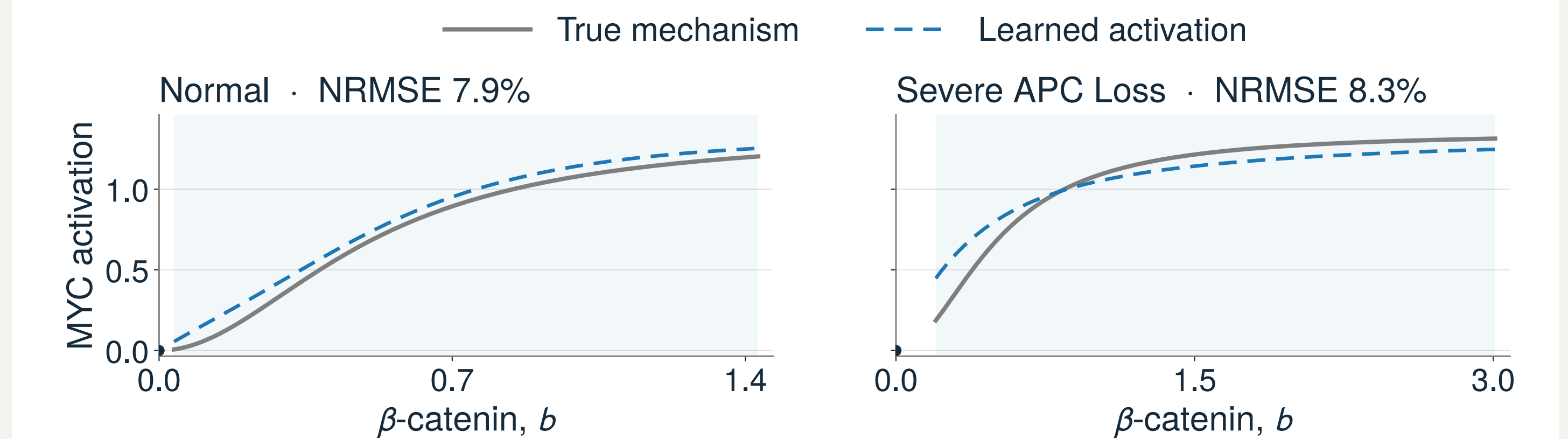


Fig. 4. Learned MYC activation versus the true mechanism, in two regimes. Dots: enforced anchor; NRMSE uses observed ranges.

Comparison with the mechanistic control

Learned term	Control	Hybrid	Function error (%)
RA $\rightarrow$ HOXA5	51/136	53/136	7.6–23.7
β -catenin $\rightarrow$ MYC	48/136	43/136	6.2–9.5
APC loss	52/140	53/140	0.12

Table 2. Parameters within 10% of truth, summed over four regimes on the common remaining parameter set. Function error: NRMSE. APC loss is calibrated first, then frozen.

Function and parameter recovery differ. MYC activation error stays below 10%, while parameter recovery falls from 48 to 43 on the shared set.

Why an unvisited anchor can fail

On the observed range, basal production and neural activation can compensate:

$$a_M + f_\phi(b) = (a_M + c) + [f_\phi(b) - c]. \quad (7)$$

13.8–202.8% basal error across five constraint forms in the Normal-regime screen.

Designing conditions that reach the anchor

Intervention	Normal	Severe
<i>MYC equation: basal a_M error (%)</i>		
WNT depletion only	2.170	1.550
WNT + HOXA13 feedback suppression	0.021	0.006
<i>Prospective HOXA13 test: basal a_{13} error (%)</i>		
Baseline	12.844	95.564
β -catenin depletion	51.516	59.843
MYC depletion	0.111	0.049

Table 3. MYC comparison: 11 conditions per arm. Prospective test: 10 baseline conditions, 11 with depletion. Strongest depletion is shown.

Reach the anchor; keep the basal term active. Counterexample: switching off W reaches the HOXA13 anchor but leaves W error at 16.2% (Normal).